Project Title: **Development of a Web and Mobile-Based Road Pothole Reporting and Monitoring System for Local Government Units**
Section: BSIT 3H G1 / BSIT 3H G2
Grouping: Minimum 4 members, Maximum 5 members
Period Timeline: October 12 – 30, 2026
Submission Date: November 02, 2026

1. Project Overview

The Web- and Mobile-Based Road Potholes Reporting System is a software application designed to help citizens report road potholes and to enable authorized government personnel, such as local government units (LGUs) and road maintenance teams, to monitor, verify, prioritize, and resolve reported road damage.

The system will provide a web-based administrative dashboard and a mobile-friendly interface for citizens to submit pothole reports using photos, GPS coordinates, descriptions, and location details. It will improve communication between citizens and road maintenance authorities, promote transparency, and help prioritize road repairs.

2. Project Objectives

The system aims to:

1. Provide a convenient way for citizens to report road potholes.
2. Capture the location of potholes using GPS or manual map selection.
3. Allow users to upload photos as evidence of road damage.
4. Enable administrators to review, verify, and manage submitted reports.
5. Track the progress of pothole repairs from submission to completion.
6. Notify users of status updates regarding their reports.
7. Provide maps, statistics, and reports to support road maintenance planning.
8. Reduce duplicate reports and improve the efficiency of road maintenance operations.

3. System Users and Roles

1. Citizen / Reporter
   - Register and log in to the system, if required.
   - Submit pothole reports with photos and location information.
   - View submitted reports and their current status.
   - Receive notifications about report progress.
   - View public pothole reports on a map.
2. Administrator / LGU Personnel
   - Access the administrative dashboard.
   - Review, verify, reject, or flag reports.
   - Assign verified reports to maintenance personnel.
   - Update report statuses.
   - Manage user accounts and system settings.
   - Generate summary reports and statistics.
3. Road Maintenance Personnel
   - View assigned pothole reports.
   - Check the location and severity of road damage.
   - Update repair progress.
   - Upload before-and-after photos.
   - Mark repair work as completed for administrative verification.

4. Functional Requirements

Functional requirements describe what the system must be able to do.

ID	Requirement	Description
FR-01	User Registration and Login	Allow users to create accounts and securely log in, where authentication is required.
FR-02	Pothole Reporting	Allow citizens to submit pothole reports through a web browser or mobile device.
FR-03	Photo Upload	Allow users to upload one or more images showing the pothole.
FR-04	GPS Location	Automatically capture coordinates when permission is granted.

FR-05 Manual Location Selection	Allow users to pin the pothole location on a map or enter an address.
FR-06 Report Description	Allow users to describe the pothole, road name, landmarks, and observed hazards.
FR-07 Severity Classification	Categorize potholes by severity, such as low, medium, or high.
FR-08 Report Management	Allow administrators to view, search, filter, verify, and manage reports.
FR-09 Status Tracking	Display the current status of each report.
FR-10 Assignment Management	Allow administrators to assign verified reports to maintenance teams.
FR-11 Notifications	Notify relevant users when report statuses change.
FR-12 Interactive Map	Display reported potholes on a map using their geographic coordinates.
FR-13 Repair Documentation	Allow maintenance personnel to submit repair notes and completion photos.
FR-14 Dashboard and Analytics	Display report totals, pending cases, completed repairs, and other relevant statistics.
FR-15 Report History	Maintain a history of report updates and actions.
FR-16 Duplicate Detection	Identify possible duplicate reports based on proximity, location, and report details.
FR-17 Feedback	Allow reporters to provide feedback on completed repairs.
FR-18 Report Export	Allow authorized personnel to export reports in formats such as CSV or PDF.

5. Pothole Reporting Workflow

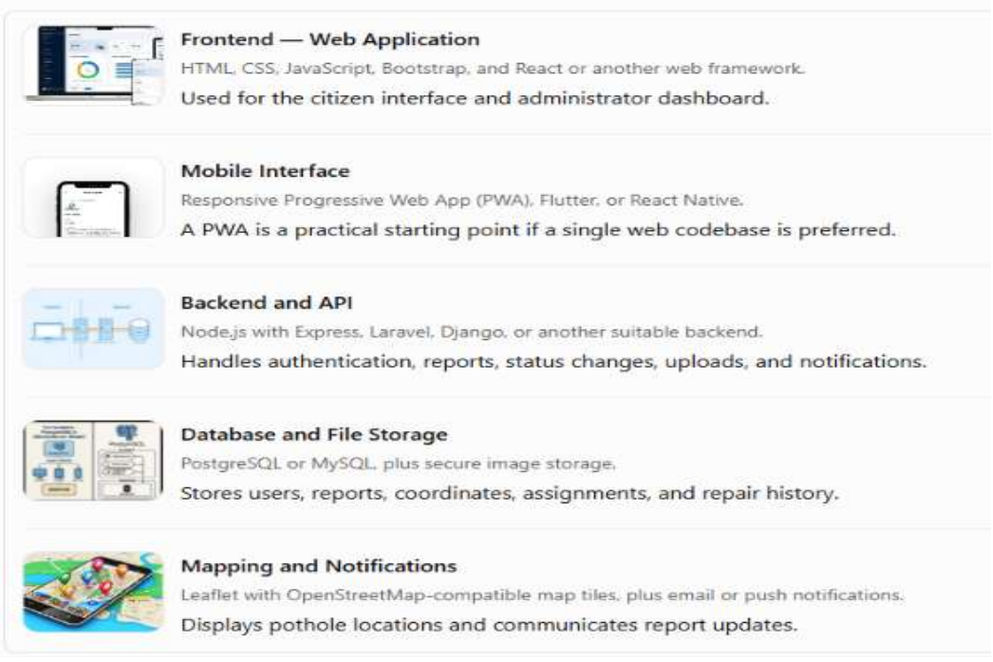
1. Citizen submits a report (Photo, location, description, and severity)
2. System validates the report (Check required fields and possible duplicates)
3. Administrator reviews the report (Verify location, evidence, and reported severity)
4. Maintenance team is assigned (Inspect the site and perform repairs)
5. Repair completion is documented (Upload completion photos and repair notes)
6. Report is verified and closed (Update the citizen and record the final status)

Recommended report statuses

- Submitted — The report has been received.
- Under Review — An administrator is checking the report.
- Verified — The pothole report has been confirmed.
- Assigned — A maintenance team has been assigned.
- In Progress — Inspection or repair work has started.
- Resolved — Repairs have been completed and verified.
- Rejected — The report is invalid or outside the responsible agency's jurisdiction.
- Duplicate — The report refers to an existing case.

6. Technical Requirements

- The following is one practical technology stack for a student or local government project. Equivalent technologies may be used if they meet the requirements.



7. User Interface Requirements

The system should contain the following screens.



A. Citizen Mobile Interface

- Home screen and report submission button
- Pothole reporting form
- Camera or image upload control
- GPS and map location selector
- My Reports page
- Report details and status timeline
- Notifications and feedback



B. Administrator Dashboard

- Total reports and pending cases
- Reports by status and severity
- Interactive pothole map
- Searchable reports table
- Report verification and assignment controls
- User management
- Statistics and export tools



C. Maintenance Personnel Interface

- Assigned repair tasks
- Pothole location and description
- Navigation or map link
- Repair progress updates
- Before-and-after photo upload
- Completion submission

8. System Constraints and Assumptions

1. GPS functionality depends on device hardware, browser permissions, and signal availability.
2. Users may manually select a location when GPS is unavailable or inaccurate.
3. The responsible LGU or road agency must have authorized personnel to review and process reports.
4. Internet access is required for real-time synchronization unless offline reporting is specifically implemented.
5. The system should identify the agency responsible for each road or provide a way to forward reports to the appropriate authority.
6. Publicly displayed reports should not expose unnecessary personal information.
7. Reports should not automatically be considered verified solely because a citizen submitted them.

9. Expected Project Outcome

The expected outcome is a working web and mobile-based pothole reporting system that connects citizens with road maintenance authorities. It will provide a centralized database of reported road damage, location-based monitoring, report status tracking, repair documentation, and administrative analytics.